Supplementary Figures

Figure S1: Population genomics of all samples. ADMIXTURE plot (left) and PCA (right) show unusual patterns of clustering between UWBM77548 and UWBM77718 that are consistent with these two individuals being close relatives. They were both sampled from the same location.

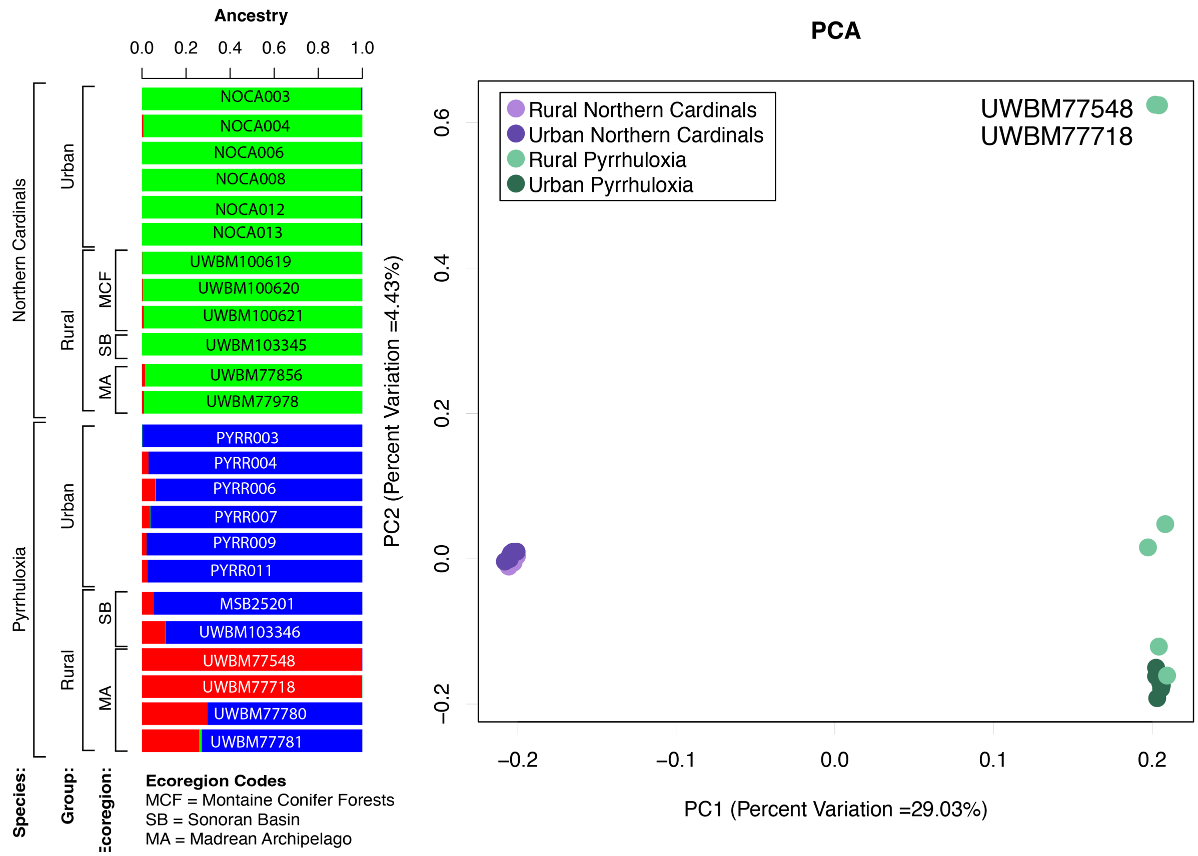
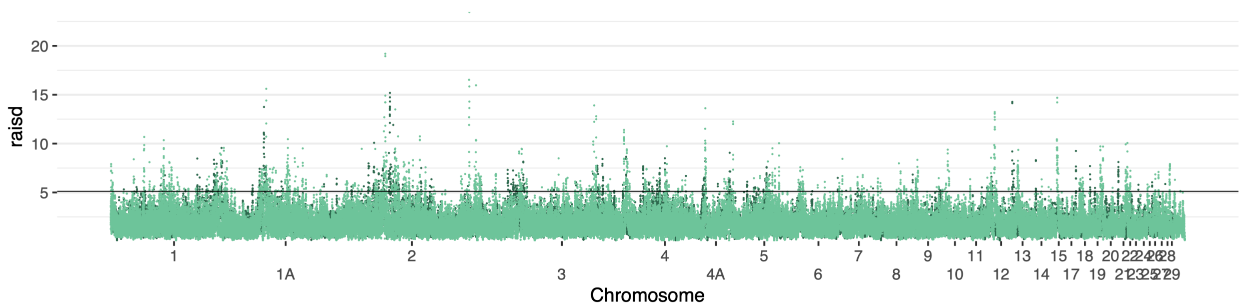
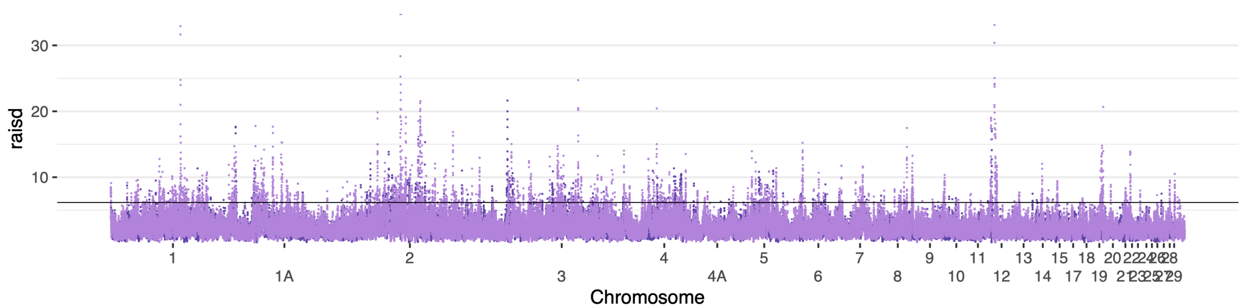


Figure S2. Signals of selection from RAiSD in pyrrhuloxia (top, green) and northern cardinals (bottom, purple). In these plots, only signals from urban samples in the absence of a signal from rural samples were considered significant. To visualize this, the rural statistics are plotted over top of the urban statistics, and only regions where the urban datapoints are visible were significant. Horizontal lines show the 1% cutoff of significance.

Pyrrhuloxia



Northern cardinals



Supplementary Tables

Table S1: Interspecific Weir and Cockerham's F_{ST} values

Pop1	Pop2	Unweighted F_{ST}	Weighted F_{ST}
Urban pyrrhuloxia	Urban northern cardinals	0.414	0.638
Urban pyrrhuloxia	Rural northern cardinals	0.415	0.638
Rural pyrrhuloxia	Urban northern cardinals	0.416	0.638
Rural pyrrhuloxia	Rural northern cardinals	0.416	0.637

Table S2: Significantly overrepresented GO terms in candidate gene lists from northern cardinals, pyrrhuloxia, or from more than one species. "Intersection" refers to the list of candidate genes identified in more than one taxon.

Species	Gene List	PANTHER GO Slim Biological Process	Log_2 Fold Enrichment	p-value	q-value
Northern cardinals	**F_{ST} and RAiSD**	platelet aggregation (GO:0070527)	18.86	0.000	0.016
		protein activation cascade (GO:0072376)	18.86	0.000	0.016
		negative regulation of blood coagulation (GO:0030195)	17.68	0.000	0.064
		platelet activation (GO:0030168)	15.72	0.000	0.022
		negative regulation of coagulation (GO:0050819)	14.15	0.001	0.106
		homotypic cell-cell adhesion (GO:0034109)	13.47	0.000	0.035
		hemostasis (GO:0007599)	10.48	0.000	0.064
		blood coagulation (GO:0007596)	10.48	0.000	0.064
		coagulation (GO:0050817)	10.48	0.000	0.064
		negative regulation of response to external stimulus (GO:0032102)	8.84	0.000	0.022
		skeletal muscle tissue development (GO:0007519)	8.42	0.000	0.058
		skeletal muscle organ development (GO:0060538)	7.37	0.000	0.069
		muscle organ development (GO:0007517)	5.89	0.001	0.159
		muscle tissue development (GO:0060537)	4.56	0.002	0.194
		cell junction assembly (GO:0034329)	3.54	0.001	0.123
		cell-cell adhesion via plasma-membrane adhesion molecules (GO:0098742)	3.37	0.001	0.106
		cell-cell adhesion (GO:0098609)	2.78	0.000	0.022
		cell adhesion (GO:0007155)	2.14	0.001	0.159
	F_{ST}	skeletal muscle tissue development (GO:0007519)	12.77	0.000	0.028
		skeletal muscle organ development (GO:0060538)	11.18	0.000	0.039
		muscle organ development (GO:0007517)	8.94	0.000	0.064
		adherens junction organization (GO:0034332)	7.77	0.000	0.085
		cell-cell adhesion via plasma-membrane adhesion molecules (GO:0098742)	5.11	0.000	0.028
		cell junction assembly (GO:0034329)	4.77	0.000	0.064
		cell junction organization (GO:0034330)	3.64	0.000	0.064

		cell-cell adhesion (GO:0098609)	3.11	0.000	0.064
		cell morphogenesis (GO:0000902)	2.87	0.000	0.064
		cell-cell signaling (GO:0007267)	2.46	0.001	0.122
		organonitrogen compound metabolic process (GO:1901564)	0.56	0.001	0.124
	RAiSD	platelet aggregation (GO:0070527)	50.93	0.000	0.000
		protein activation cascade (GO:0072376)	50.93	0.000	0.000
		negative regulation of blood coagulation (GO:0030195)	47.75	0.000	0.003
		platelet activation (GO:0030168)	42.44	0.000	0.001
		negative regulation of coagulation (GO:0050819)	38.2	0.000	0.008
		homotypic cell-cell adhesion (GO:0034109)	36.38	0.000	0.001
		hemostasis (GO:0007599)	28.29	0.000	0.002
		blood coagulation (GO:0007596)	28.29	0.000	0.002
		coagulation (GO:0050817)	28.29	0.000	0.002
		regulation of response to wounding (GO:1903034)	27.28	0.000	0.018
		calcineurin-mediated signaling (GO:0097720)	27.28	0.000	0.018
		regulation of wound healing (GO:0061041)	27.28	0.000	0.018
		regulation of coagulation (GO:0050818)	23.87	0.000	0.025
		negative regulation of response to external stimulus (GO:0032102)	19.89	0.000	0.001
		positive regulation of cell-cell adhesion (GO:0022409)	17.36	0.001	0.060
		mismatch repair (GO:0006298)	17.36	0.001	0.060
		regulation of body fluid levels (GO:0050878)	16.98	0.000	0.013
		wound healing (GO:0042060)	13.4	0.000	0.025
		response to wounding (GO:0009611)	10.19	0.001	0.060
		cell activation (GO:0001775)	8.78	0.001	0.093
		regulation of response to external stimulus (GO:0032101)	7.64	0.000	0.018
		tRNA processing (GO:0008033)	5.58	0.002	0.170
		regulation of response to stress (GO:0080134)	5.3	0.001	0.087
		tRNA metabolic process (GO:0006399)	4.79	0.001	0.064
Pyrrhuloxia	**F_{ST} and RAiSD**	None	None	None	None
	F_{ST}	secretory granule organization (GO:0033363)	35.66	0.001	0.096
		regeneration (GO:0031099)	35.66	0.001	0.096
		tissue regeneration (GO:0042246)	35.66	0.001	0.096

		insulin-like growth factor receptor signaling pathway (GO:0048009)	21.4	0.000	0.060
		regulation of phosphoprotein phosphatase activity (GO:0043666)	21.4	0.000	0.060
		regulation of phosphatase activity (GO:0010921)	17.83	0.000	0.075
		regulation of dephosphorylation (GO:0035303)	11.89	0.002	0.153
		nervous system development (GO:0007399)	2.09	0.000	0.075
		metabolic process (GO:0008152)	0.66	0.000	0.060
		nitrogen compound metabolic process (GO:0006807)	0.64	0.001	0.092
		organic substance metabolic process (GO:0071704)	0.64	0.000	0.060
		primary metabolic process (GO:0044238)	0.63	0.000	0.060
		cellular metabolic process (GO:0044237)	0.62	0.000	0.075
		macromolecule biosynthetic process (GO:0009059)	0.44	0.001	0.115
		heterocycle metabolic process (GO:0046483)	0.44	0.001	0.115
		cellular aromatic compound metabolic process (GO:0006725)	0.44	0.001	0.115
		nucleobase-containing compound metabolic process (GO:0006139)	0.43	0.001	0.116
		organic cyclic compound metabolic process (GO:1901360)	0.42	0.000	0.075
		cellular nitrogen compound metabolic process (GO:0034641)	0.42	0.000	0.060
		organic substance biosynthetic process (GO:1901576)	0.42	0.000	0.036
		biosynthetic process (GO:0009058)	0.42	0.000	0.036
		cellular biosynthetic process (GO:0044249)	0.4	0.000	0.036
		regeneration (GO:0031099)	35.47	0.001	0.183
		tissue regeneration (GO:0042246)	35.47	0.001	0.183
		regulation of centrosome duplication (GO:0010824)	17.73	0.000	0.124
		regulation of centrosome cycle (GO:0046605)	17.73	0.000	0.124
		proton transmembrane transport (GO:1902600)	12.67	0.000	0.029
		tRNA modification (GO:0006400)	6.06	0.000	0.068
		tRNA processing (GO:0008033)	5.6	0.000	0.029
		tRNA metabolic process (GO:0006399)	4.19	0.000	0.041
		RNA modification (GO:0009451)	4.04	0.000	0.124
Intersection	**All**	regulation of muscle system process (GO:0090257)	56.38	0.001	0.041

	skeletal muscle tissue development (GO:0007519)	39.03	0.001	0.075
	skeletal muscle organ development (GO:0060538)	33.83	0.002	0.095
	adherens junction organization (GO:0034332)	33.09	0.000	0.011
	homophilic cell adhesion via plasma membrane adhesion molecules (GO:0007156)	29.27	0.000	0.013
	smoothened signaling pathway (GO:0007224)	28.19	0.002	0.117
	muscle organ development (GO:0007517)	26.7	0.002	0.126
	cell-cell adhesion via plasma-membrane adhesion molecules (GO:0098742)	25.37	0.000	0.000
	synapse assembly (GO:0007416)	23.06	0.003	0.154
	cell junction assembly (GO:0034329)	21.5	0.000	0.001
	cell-cell junction assembly (GO:0007043)	20.57	0.000	0.033
	synapse organization (GO:0050808)	19.82	0.000	0.001
	cell-cell junction organization (GO:0045216)	17.7	0.001	0.047
	cell junction organization (GO:0034330)	16.75	0.000	0.000
	cell-cell adhesion (GO:0098609)	11.24	0.000	0.001
	synaptic signaling (GO:0099536)	10.08	0.000	0.004
	cell morphogenesis (GO:0000902)	9.7	0.000	0.001
	axon guidance (GO:0007411)	9.57	0.001	0.055
	neuron projection guidance (GO:0097485)	9.57	0.001	0.055
	cell adhesion (GO:0007155)	8.9	0.000	0.001
	neuron development (GO:0048666)	8.54	0.000	0.002
	axonogenesis (GO:0007409)	8.05	0.002	0.094
	cell morphogenesis involved in neuron differentiation (GO:0048667)	7.75	0.002	0.103
	chemical synaptic transmission (GO:0007268)	7.35	0.002	0.117
	anterograde trans-synaptic signaling (GO:0098916)	7.35	0.002	0.117
	axon development (GO:0061564)	7.2	0.002	0.117
	trans-synaptic signaling (GO:0099537)	7.2	0.002	0.117
	neuron projection morphogenesis (GO:0048812)	6.72	0.003	0.140
	cell projection morphogenesis (GO:0048858)	6.68	0.003	0.140

	plasma membrane bounded cell projection morphogenesis (GO:0120039)	6.68	0.003	0.140
	neuron projection development (GO:0031175)	6.61	0.001	0.061
	neuron differentiation (GO:0030182)	6.53	0.000	0.011
	cell-cell signaling (GO:0007267)	6.32	0.000	0.029
	nervous system development (GO:0007399)	6.28	0.000	0.001
	generation of neurons (GO:0048699)	6.28	0.000	0.012
	cell development (GO:0048468)	5.82	0.000	0.016
	neurogenesis (GO:0022008)	5.8	0.000	0.016
	multicellular organism development (GO:0007275)	5.72	0.000	0.000
	anatomical structure morphogenesis (GO:0009653)	5.49	0.000	0.011
	system development (GO:0048731)	5.15	0.000	0.001
	multicellular organismal process (GO:0032501)	4.52	0.000	0.000
	anatomical structure development (GO:0048856)	4.27	0.000	0.000
	developmental process (GO:0032502)	3.8	0.000	0.001
	signaling (GO:0023052)	2.9	0.000	0.011
	cell communication (GO:0007154)	2.84	0.000	0.013

Table S3: Genes associated with significant GO terms in analyses of candidate gene lists from northern cardinals, pyrrhuloxia, or from more than one species. "Intersection" refers to the list of candidate genes identified in more than one taxon.

Species	Analysis	GO_Term	Name	Genes
Northern cardinals	**F_{ST} and RAiSD**	GO:0070527	platelet aggregation	*FGA, FGB*
		GO:0072376	protein activation cascade	*FGA, FGB*
		GO:0030195	negative regulation of blood coagulation	*FGA, FGB*
		GO:0030168	platelet activation	*FGA, FGB*
		GO:0050819	negative regulation of coagulation	*FGA, FGB*
		GO:0034109	homotypic cell-cell adhesion	*FGA, FGB*
		GO:0007599	hemostasis	*FGA, FGB*
		GO:0007596	blood coagulation	*FGA, FGB*
		GO:0050817	coagulation	*FGA, FGB*
		GO:0032102	negative regulation of response to external stimulus	*SUSD4, FGA, FGB, CACTIN, PPARG*
		GO:0007519	skeletal muscle tissue development	*MYF5, MYF6, DMD*
		GO:0060538	skeletal muscle organ development	*MYF5, MYF6, DMD*
		GO:0007517	muscle organ development	*MYF5, MYF6, DMD*
		GO:0060537	muscle tissue development	*MYF5, MYF6, DMD, SGCZ*
		GO:0034329	cell junction assembly	*CDH11, CDH18, SDK1, CDH20, CLDN23, NLGN3, CDH9*
		GO:0098742	cell-cell adhesion via plasma-membrane adhesion molecules	*CDH11, ATRN, CDH18, SDK1, CNTN4, CDH20, MEGF11*
		GO:0098609	cell-cell adhesion	*DCHS1, FGA, CDH11, ATRN, CDH18, SDK1, CNTN4, CDH11, CDH20, FGB, MEGF11, TJP3, SDK1, NLGN3, CDON, ITGA9, FGA, CDH9*
		GO:0007155	cell adhesion	*DCHS1, FGA, CDH11, ATRN, CDH18, SDK1, CNTN4, CDH11, CDH20, FGB, MEGF11, TJP3, SDK1, NLGN3, CDON, ITGA9, FGA, CDH9, PCDH15, CLDN23*
	F_{ST}	GO:0007519	skeletal muscle tissue development	*MYF5, MYF6, DMD*
		GO:0060538	skeletal muscle organ development	*MYF5, MYF6, DMD*
		GO:0007517	muscle organ development	*MYF5, MYF6, DMD*

	GO:0034332	adherens junction organization	*CDH11, CDH18, CDH20, CDH9*
	GO:0098742	cell-cell adhesion via plasma-membrane adhesion molecules	*CDH11, ATRN, CDH18, SDK1, CNTN4, CDH11, CDH20, MEGF11, SDK1, CDH9*
	GO:0034329	cell junction assembly	*CDH11, CDH18, SDK1, CDH11, CDH20, NLGN3, CDH9*
	GO:0034330	cell junction organization	*CDH11, CDH18, SDK1, CNTN4, CDH11, CDH20, TJP3, SDK1, NLGN3, ERC2, CDH9*
	GO:0098609	cell-cell adhesion	*DCHS1, CDH11, ATRN, CDH18, SDK1, CNTN4, CDH11, CDH20, MEGF11, TJP3, SDK1, NLGN3, CDON, CDH9*
	GO:0000902	cell morphogenesis	*UNC5C, SLITRK1, EPHA10, CDH11, TRIO, EPHB2, CDH18, SLIT3, UNCSC, CNTN4, CDH20, FRY, CDH9*
	GO:0007267	cell-cell signaling	*GNAI2, BCL9, LRP2, GJB1, SYN2, GRIA2, GJA6, GLRB, FCHSD2, GRID, HTR7, DMD, GJA5, NLGN3, ERC2*
	GO:1901564	organonitrogen compound metabolic process	*STT3A, TRIM2, EPHA10, CHPF, SRBD1, CERS6, EPHB2, ZDHHC20, TG, VRK1, AGBL1, MRPL57, EIF2AK4, MAP4K3, GSTZ1, ENPP2, AK4, FBXO32, DPY19L3, EPHB2, RPS6KA2, EEFSEC, PSMD14, HECW1, ANKZF1, CHEK1, MGAT5B*
RAiSD	GO:0070527	platelet aggregation	*FGA, FGB*
	GO:0072376	protein activation cascade	*FGA, FGB*
	GO:0030195	negative regulation of blood coagulation	*FGA, FGB*
	GO:0030168	platelet activation	*FGA, FGB*
	GO:0050819	negative regulation of coagulation	*FGA, FGB*
	GO:0034109	homotypic cell-cell adhesion	*FGA, FGB*
	GO:0007599	hemostasis	*FGA, FGB*
	GO:0007596	blood coagulation	*FGA, FGB*
	GO:0050817	coagulation	*FGA, FGB*

		GO:1903034	regulation of response to wounding	*FGA, FGB*
		GO:0097720	**calcineurin-mediated signaling**	*PPP3R1, PPP3CA, NFATC1*
		GO:0061041	regulation of wound healing	*FGA, FGB*
		GO:0050818	regulation of coagulation	*FGA, FGB*
		GO:0032102	negative regulation of response to external stimulus	*FGA, FGB, SUSD4, PPARG*
		GO:0022409	**positive regulation of cell-cell adhesion**	*FGA, FGB*
		GO:0006298	**mismatch repair**	*MSH2, MSH6*
		GO:0050878	**regulation of body fluid levels**	*FGA, FGB*
		GO:0042060	wound healing	*FGA, FGB*
		GO:0009611	response to wounding	*FGA, FGB*
		GO:0001775	cell activation	*FGA, FGB*
		GO:0032101	regulation of response to external stimulus	*FGA, FGB, SUSD4, PPARG, TLR5*
		GO:0008033	**tRNA processing**	*TSEN2, TRMT61B, TRMT9B, CDKAL1, TSEN2*
		GO:0080134	**regulation of response to stress**	*FGA, FGB, SUSD4, PPARG, TLR5*
		GO:0006399	**tRNA metabolic process**	*TSEN2, TRMT61B, TRMT9B, CDKAL1, TSEN2, GATB, DTD1*
Pyrrhuloxia	**F_{ST} and RAiSD**	None	None	*None*
	F_{ST}	GO:0033363	secretory granule organization	*DTNBP1*
		GO:0031099	regeneration	*GAP43*
		GO:0042246	tissue regeneration	*GAP43*
		GO:0048009	insulin-like growth factor receptor signaling pathway	*GIGYF2, IGF1*
		GO:0043666	regulation of phosphoprotein phosphatase activity	*PPP6R2, PPP6R3*
		GO:0010921	regulation of phosphatase activity	*PPP6R2, PPP6R3*
		GO:0035303	regulation of dephosphorylation	*PPP6R2, PPP6R3*
		GO:0007399	nervous system development	*RUNX2, GPM6A, EPHA10, DTNBP1, ABI2, WNT16, GAP43, RBFOX1, IGF2BP2, DTNBP1, SDK1, SLIT3, NAV2, NTN4, TENM3, TRAK1, KIRREL3, SDK1, ERBB4, DMD, NYAP2, NLGN3, SEMA6D*
		GO:0008152	metabolic process	*GGT7, STT3A, SUMO2, PDPK1, CSGALNACT1, UBA6, TRIP12, EPHA10, TASP1, MTMR4, NCF4, PLCE1, HEXA, CDKAL1, RAB19, RNGTT, EI24,*

			OTUD7A, STK4, PBRM1, CTSC, DROSHA, G6PC2, UBE2I, DCUN1D1, CACTIN, RNF43, EIF3F, CDK14, NUP85
	GO:0006807	nitrogen compound metabolic process	*GGT7, STT3A, SUMO2, PDPK1, CSGALNACT1, UBA6, TRIP12, EPHA10, TASP1, HEXA, CDKAL1, RNGTT, OTUD7A, STK4, PBRM1, CTSC, DROSHA, UBE2I, DCUN1D1, CACTIN, RNF43, EIF3F, CDK14, KYNU, MAN1A2, NOC3L, RNF13, ARIH1, DYRK1A, PRKCH, LOXL4, ANKIB1, PBRM1, TDO2, CTSO, STK4, MTA1, FUT8, BABAM2, PPP2R3B, SENP2, ENTPD1, CHEK1, SUMO2*
	GO:0071704	organic substance metabolic process	*GGT7, STT3A, SUMO2, PDPK1, CSGALNACT1, UBA6, TRIP12, EPHA10, TASP1, MTMR4, PLCE1, HEXA, CDKAL1, RNGTT, OTUD7A, STK4, PBRM1, CTSC, DROSHA, G6PC2, UBE2I, DCUN1D1, CACTIN, RNF43, EIF3F, CDK14, NUP85, KYNU, MAN1A2, NOC3L, RNF13, NDUFA9, ARIH1, DYRK1A, PRKCH, LOXL4, PBRM1, ANKIB1, TDO2, CTSO, STK4, MTA1, FUT8, BABAM2, PIP5K1C, PPP2R3B, SENP2, ENTPD1, CHEK1, SUMO2*
	GO:0044238	primary metabolic process	*STT3A, SUMO2, PDPK1, CSGALNACT1, UBA6, TRIP12, EPHA10, TASP1, MTMR4, PLCE1, HEXA, CDKAL1, RNGTT, OTUD7A, STK4, PBRM1, CTSC, DROSHA, G6PC2, UBE2I, DCUN1D1, CACTIN, RNF43, EIF3F, CDK14, KYNU, MAN1A2, NOC3L, RNF13, ARIH1, DYRK1A, PRKCH, LOXL4, ANKIB1, PBRM1, TDO2, CTSO, STK4, MTA1, FUT8, BABAM2, PIP5K1C, PPP2R3B, SENP2, ENTPD1, CHEK1, SUMO2*

	GO:0044237	cellular metabolic process	*GGT7, STT3A, PDPK1, CSGALNACT1, EPHA10, TASP1, MTMR4, NCF4, PLCE1, HEXA, CDKAL1, RAB19, RNGTT, EI24, STK4, PBRM1, DROSHA, G6PC2, CACTIN, RNF43, EIF3F, CDK14, NUP85, KYNU, NOC3L, NDUFA9, DYRK1A, PRKCH, PBRM1, TDO2, STK4, MTA1, FUT8, SNX7, BABAM2, PIP5K1C, PPP2R3B, ENTPD1, CHEK1*
	GO:0009059	macromolecule biosynthetic process	*STT3A, CSGALNACT1, TASP1, CDKAL1, RNGTT, PBRM1, DROSHA, CACTIN, EIF3F, NUP85, PBRM1, FUT8*
	GO:0046483	heterocycle metabolic process	*CDKAL1, RNGTT, PBRM1, DROSHA, CACTIN, KYNU, NOC3L, PBRM1, TDO2, MTA1, BABAM2, ENTPD1*
	GO:0006725	cellular aromatic compound metabolic process	*CDKAL1, RNGTT, PBRM1, DROSHA, CACTIN, KYNU, NOC3L, PBRM1, TDO2, MTA1, BABAM2, ENTPD1*
	GO:0006139	nucleobase-containing compound metabolic process	*CDKAL1, RNGTT, PBRM1, DROSHA, CACTIN, NOC3L, PBRM1, TDO2, MTA1, BABAM2, ENTPD1*
	GO:1901360	organic cyclic compound metabolic process	*CDKAL1, RNGTT, PBRM1, DROSHA, CACTIN, KYNU, NOC3L, PBRM1, TDO2, MTA1, BABAM2, ENTPD1*
	GO:0034641	cellular nitrogen compound metabolic process	*CDKAL1, RNGTT, PBRM1, DROSHA, CACTIN, EIF3F, KYNU, NOC3L, PBRM1, TDO2, MTA1, BABAM2, ENTPD1*
	GO:1901576	organic substance biosynthetic process	*STT3A, CSGALNACT1, TASP1, CDKAL1, RNGTT, PBRM1, DROSHA, G6PC2, CACTIN, EIF3F, NUP85, NDUFA9, PBRM1, FUT8, PIP5K1C*
	GO:0009058	biosynthetic process	*STT3A, CSGALNACT1, TASP1, CDKAL1, RNGTT, PBRM1, DROSHA, G6PC2, CACTIN, EIF3F, NUP85, NDUFA9, PBRM1, FUT8, PIP5K1C*
	GO:0044249	cellular biosynthetic process	*STT3A, CSGALNACT1, TASP1, CDKAL1, RNGTT, PBRM1, DROSHA,*

				CACTIN, EIF3F, NUP85, NDUFA9, PBRM1, FUT8, PIP5K1C
	RAiSD	GO:0031099	regeneration	*GAP43*
		GO:0042246	tissue regeneration	*GAP43*
		GO:0010824	regulation of centrosome duplication	*NPM1, SPICE1*
		GO:0046605	regulation of centrosome cycle	*NPM1, SPICE1*
		GO:1902600	proton transmembrane transport	*ATP1A1, OTOP3, OTOP2, ATP6V1G1*
		GO:0006400	tRNA modification	*CDKAL1, MTO1, THADA, QTRT2, ELP4*
		GO:0008033	tRNA processing	*RTCB, CDKAL1, MTO1, THADA, QTRT2, TSEN54, ELP4*
		GO:0006399	tRNA metabolic process	*RTCB, CDKAL1, MTO1, THADA, QTRT2, TSEN54, ELP4, EARS2*
		GO:0009451	RNA modification	*CDKAL1, MTO1, METTL16, THADA, QTRT2, ELP4*
Intersection	**All**	GO:0090257	regulation of muscle system process	*DMD*
		GO:0007519	skeletal muscle tissue development	*DMD*
		GO:0060538	skeletal muscle organ development	*DMD*
		GO:0034332	adherens junction organization	*CDH18, CDH8, CDH9*
		GO:0007156	homophilic cell adhesion via plasma membrane adhesion molecules	*SDK1, CNTN4*
		GO:0007224	smoothened signaling pathway	*GL13*
		GO:0007517	muscle organ development	*DMD*
		GO:0098742	cell-cell adhesion via plasma-membrane adhesion molecules	*ATRN*
		GO:0007416	synapse assembly	*SDK1*
		GO:0034329	cell junction assembly	*SDK1, CDH18, CDH8, CDH9*
		GO:0007043	cell-cell junction assembly	*CDH18, CDH8, CDH9*
		GO:0050808	synapse organization	*SDK1, CNTN4, ERC2, CDH9*
		GO:0045216	cell-cell junction organization	*CDH18, CDH8, CDH9*
		GO:0034330	cell junction organization	*CDH18, SDK1, CNTN4, CDH8, ERC2, CDH9*
		GO:0098609	cell-cell adhesion	*ATRN, ADH18, SDK1, CNTN4, CDH8, CDH9*
		GO:0099536	synaptic signaling	*SYT1, HTR7, DMD, ERC2*
		GO:0000902	cell morphogenesis	*TRIO, EPHB2, CDH18, CNTN4, CDH8, CDH9*
		GO:0007411	axon guidance	*TRIO, EPHB2, CNTN4*

	GO:0097485	neuron projection guidance	*TRIO, EPHB2, CNTN4*
	GO:0007155	cell adhesion	*PCDH15, ATRN, CDH18, SDK1, CNTN4, CDH8, CDH9*
	GO:0048666	neuron development	*GPM6A, TRIO, EPHB2, CNTN4, DMD*
	GO:0007409	axonogenesis	*TRIO,EPHB2, CNTN4*
	GO:0048667	cell morphogenesis involved in neuron differentiation	*TRIO, EPHB2, CNTN4*
	GO:0007268	chemical synaptic transmission	*SYT1, HTR7, ERC2*
	GO:0098916	anterograde trans-synaptic signaling	*SYT1, HTR7, ERC2*
	GO:0061564	axon development	*TRIO, EPHB2, CNTN4*
	GO:0099537	trans-synaptic signaling	*SYT1, HTR7, ERC2*
	GO:0048812	neuron projection morphogenesis	*TRIO, EPHB2, CNTN4*
	GO:0048858	cell projection morphogenesis	*TRIO, EPHB2, CNTN4*
	GO:0120039	plasma membrane bounded cell projection morphogenesis	*TRIO, EPHB2, CNTN4*
	GO:0031175	neuron projection development	*GPM6A, TRIO, EPHB2, CNTN4*
	GO:0030182	neuron differentiation	*GPM6A, TRIO, EPHB2, CNTN4, DMD*
	GO:0007267	cell-cell signaling	*SYT1, HTR7, DMD, ERC2*
	GO:0007399	nervous system development	*GPM6A, TRIO, EPHB2, SDK1, NAV2, CNTN4, DMD*
	GO:0048699	generation of neurons	*GPM6A, TRIO, EPHB2, CNTN4, DMD*
	GO:0048468	cell development	*GPM6A, TRIO, EPHB2, CNTN4, DMD*
	GO:0022008	neurogenesis	*GPM6A, TRIO, EPHB2, CNTN4, DMD*
	GO:0007275	multicellular organism development	*GPM6A, TRIO, EPHB2, CDH18, SDK1, APOLD1, ALK, NAV2, CNTN4, DMD, CDH8, CDH9*
	GO:0009653	anatomical structure morphogenesis	*TRIO, EPHB2, CDH18, APOLD1, CNTN4, EPHB2, CDH8, CDH9*
	GO:0048731	system development	*GPM6A, TRIO, EPHB2, SDK1, APOLD1, NAV2, CNTN4, DMD*
	GO:0032501	multicellular organismal process	*GPM6A, TRIO, EPHB2, CDH18, SDK1, APOLD1, ALK, NAV2, CNTN4, DMD, CDH8, CDH9*
	GO:0048856	anatomical structure development	*GPM6A, TRIO, EPHB2, CDH18, SDK1, APOLD1, ALK, NAV2, CNTN4, DMD, CDH8, CDH9*

	GO:0032502	*GPM6A, TRIO, EPHB2, CDH18, SDK1, APOLD1, ALK, NAV2, CNTN4, DMD, CDH8, CDH9*
	GO:0023052	*CHRM3, SYT1, BDKRB1, EPHB2, VRK1, GLI3, ALK, ANKS1B, HTR7, DMD, ERC2*
	GO:0007154	*GPM6A, TRIO, EPHB2, CDH18, SDK1, APOLD1, ALK, NAV2, CNTN4, DMD, CDH8, CDH9*

Table S4: List of all candidate genes for the four species in this study (see accompanying CSV file).